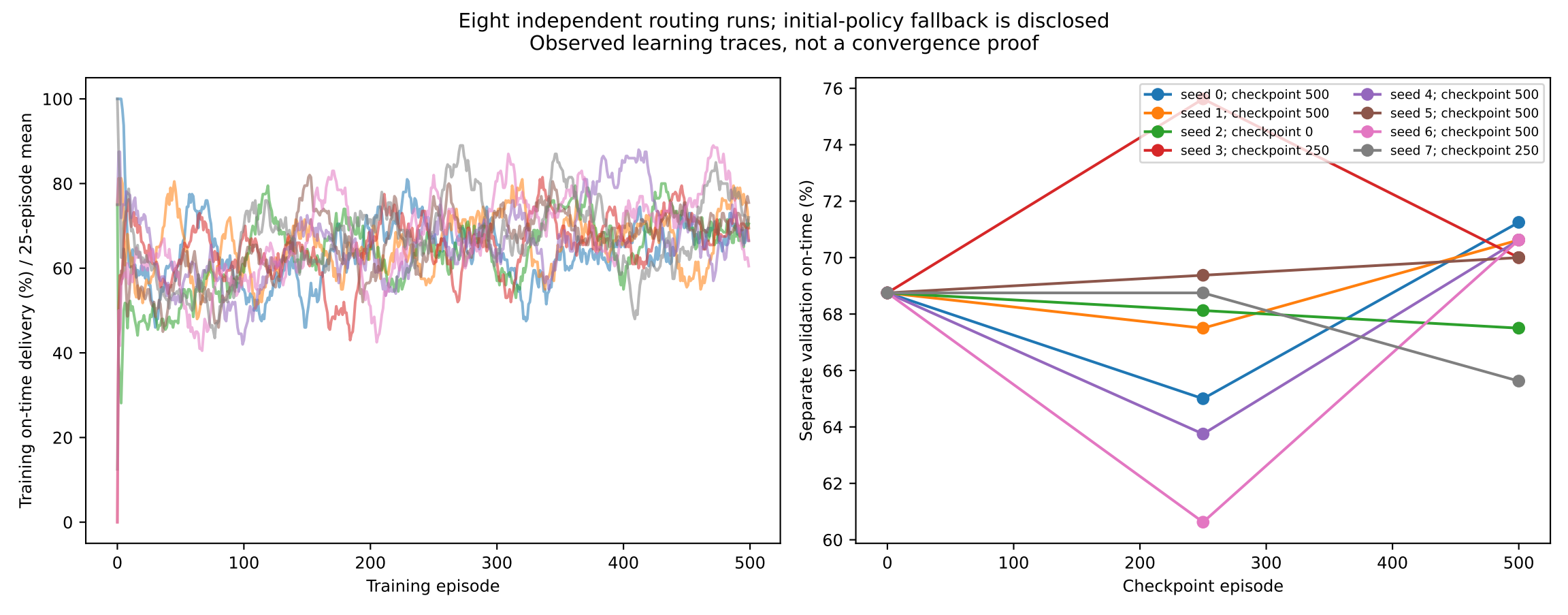
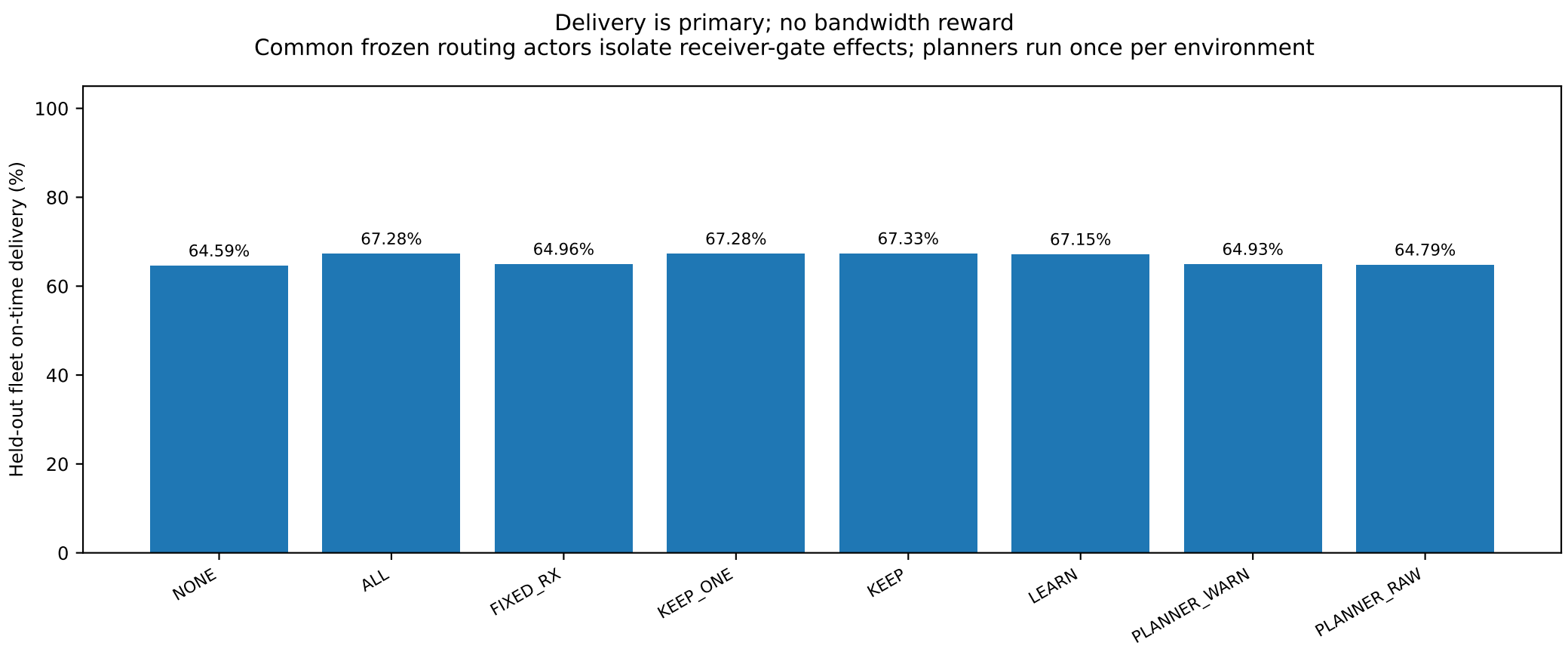
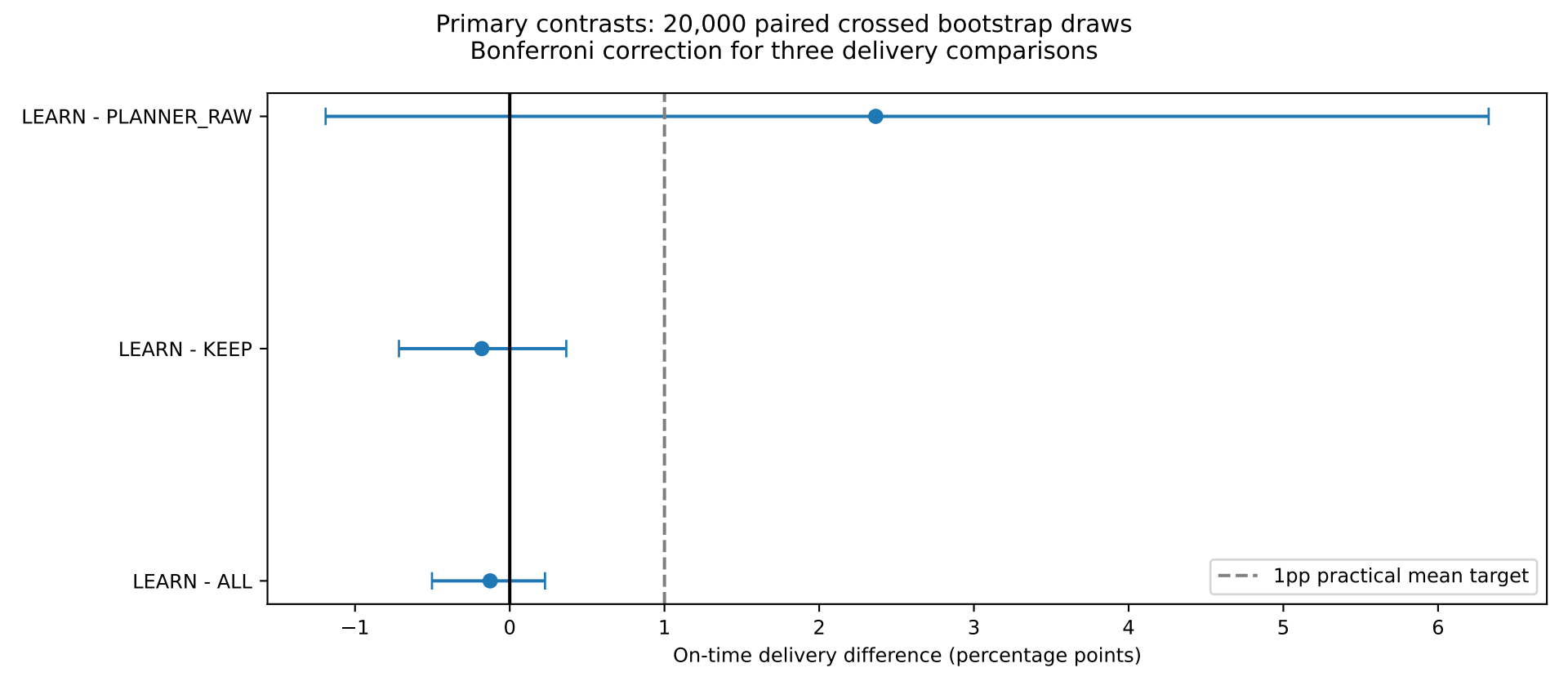
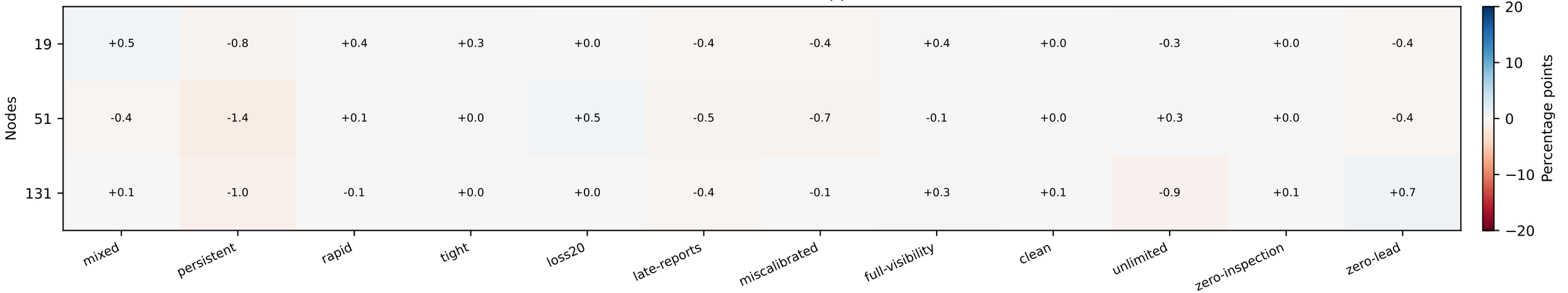

Eight independent routing runs; initial-policy fallback is disclosed
Observed learning traces, not a convergence proof
Training on-time delivery (%) / 25-episode mean
Training episode
Separate validation on-time (%)
Checkpoint episode
seed 0; checkpoint 500
seed 1; checkpoint 500
seed 2; checkpoint 0
seed 3; checkpoint 250
seed 4; checkpoint 500
seed 5; checkpoint 500
seed 6; checkpoint 500
seed 7; checkpoint 250


Delivery is primary; no bandwidth reward
Common frozen routing actors isolate receiver-gate effects; planners run once per environment
Held-out fleet on-time delivery (%)
0
20
40
60
80
100
64.59%
67.28%
64.96%
67.28%
67.33%
67.15%
64.93%
64.79%
NONE
ALL
FIXED_RX
KEEP_ONE
KEEP
LEARN
PLANNER_WARN
PLANNER_RAW

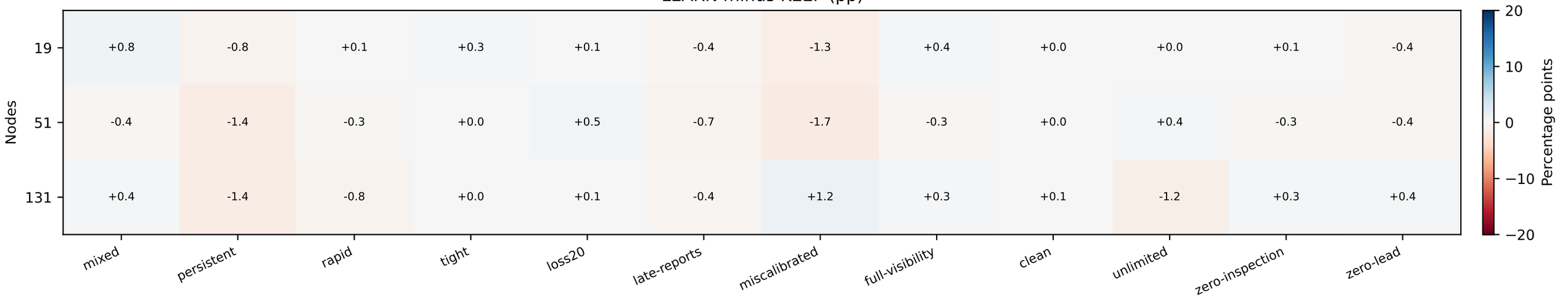

Primary contrasts: 20,000 paired crossed bootstrap draws
Bonferroni correction for three delivery comparisons
LEARN - PLANNER_RAW
LEARN - KEEP
LEARN - ALL
1pp practical mean target
−1
0
1
2
3
4
5
6
On-time delivery difference (percentage points)

Every declared condition; no favorable-case selection

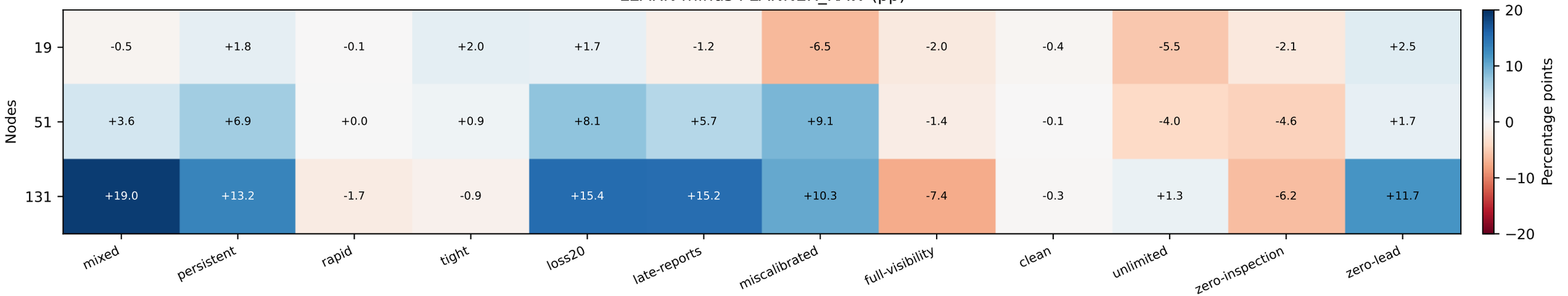
LEARN minus ALL (pp)

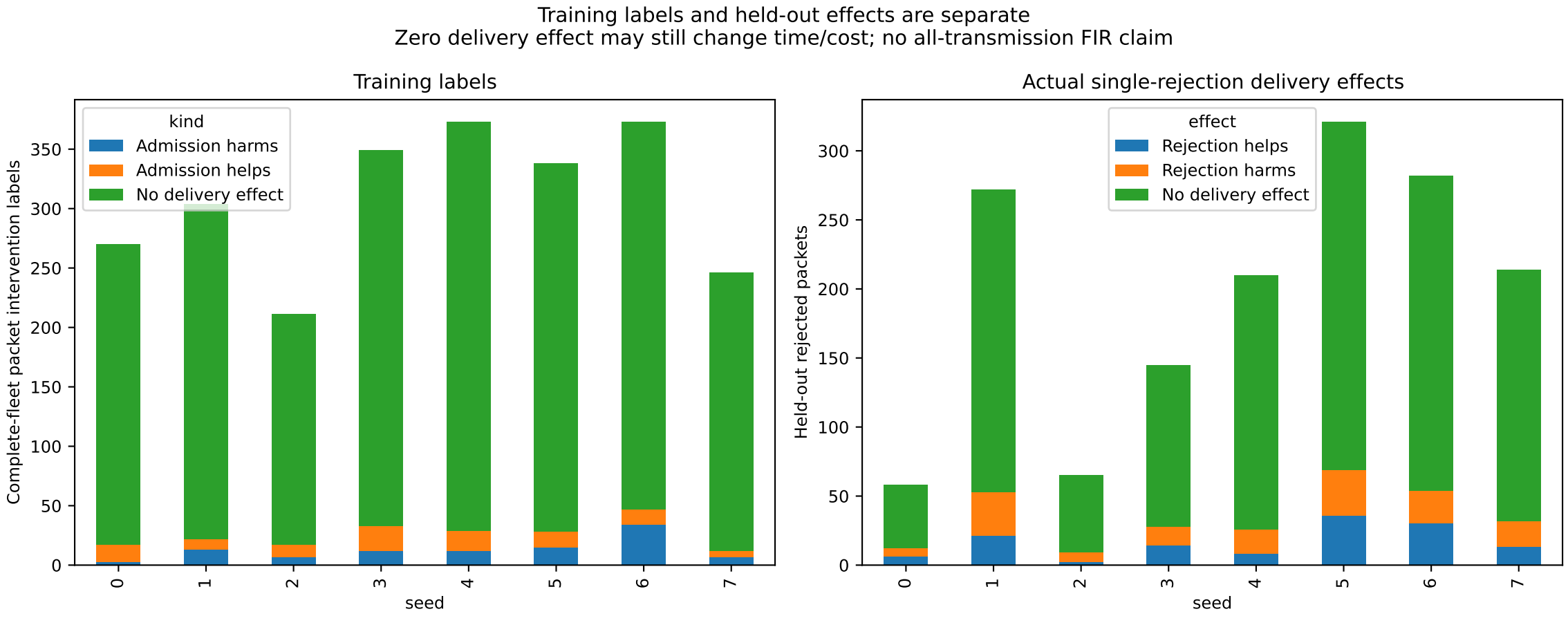


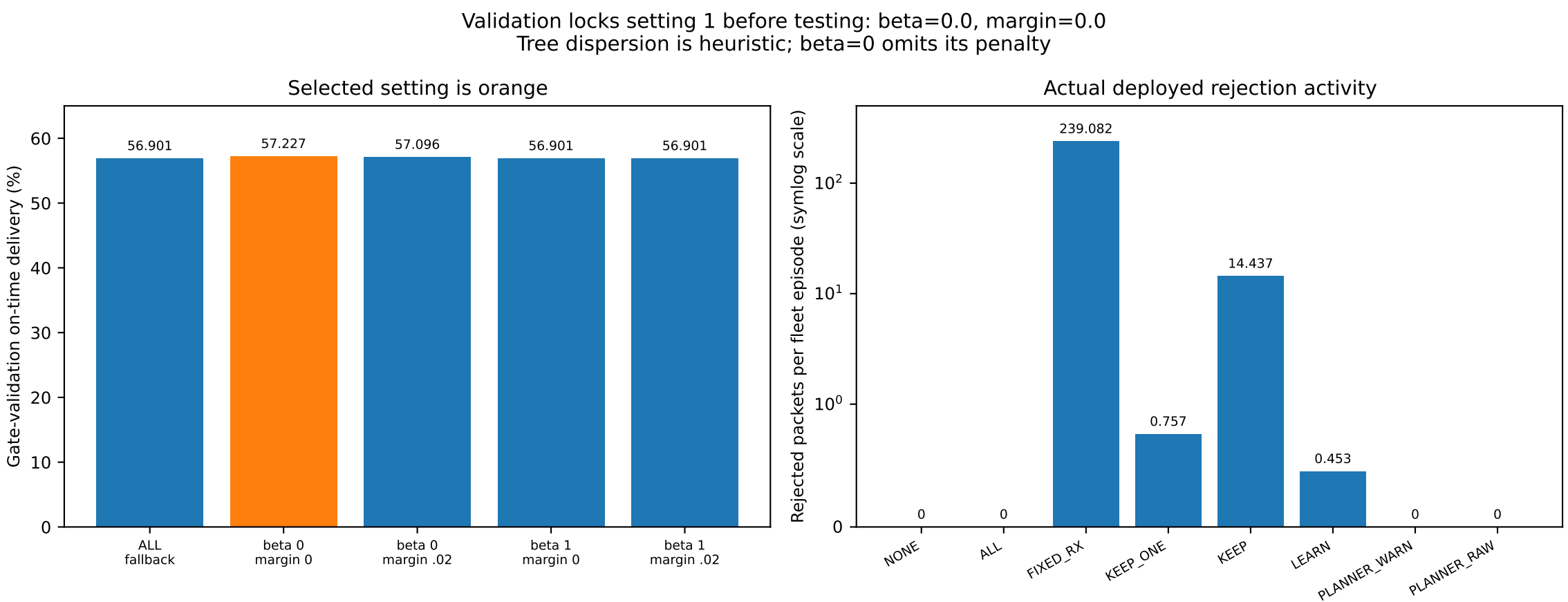
LEARN minus KEEP (pp)

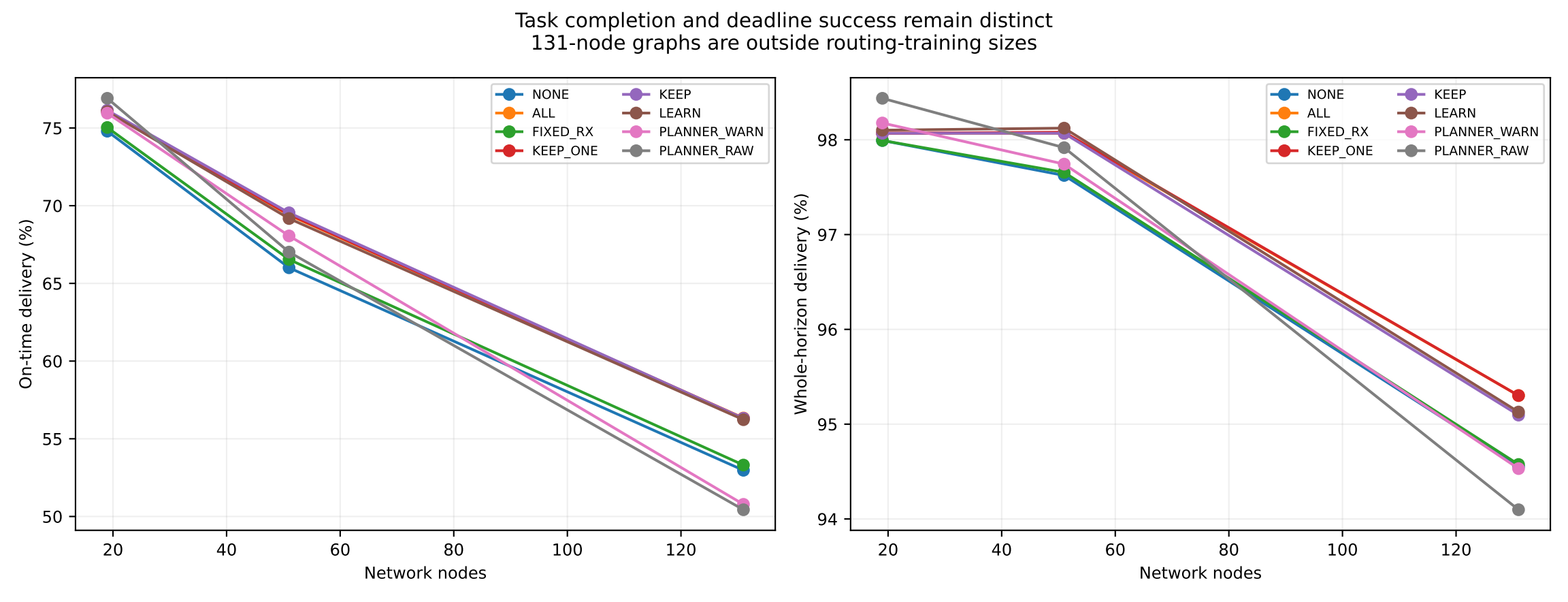


LEARN minus PLANNER_RAW (pp)




Training labels and held-out effects are separate
Zero delivery effect may still change time/cost; no all-transmission FIR claim
Training labels
Complete-fleet packet intervention labels
kind
Admission harms
Admission helps
No delivery effect
seed
Actual single-rejection delivery effects
Held-out rejected packets
effect
Rejection helps
Rejection harms
No delivery effect
seed


Validation locks setting 1 before testing: beta=0.0, margin=0.0
Tree dispersion is heuristic; beta=0 omits its penalty
Selected setting is orange
Gate-validation on-time delivery (%)
0
10
20
30
40
50
60
56.901
57.227
57.096
56.901
56.901
ALL fallback
beta 0 margin 0
beta 0 margin .02
beta 1 margin 0
beta 1 margin .02
Actual deployed rejection activity
Rejected packets per fleet episode (symlog scale)
0
10^0
10^1
10^2
0
0
239.082
0.757
14.437
0.453
0
0
NONE
ALL
FIXED_RX
KEEP_ONE
KEEP
LEARN
PLANNER_WARN
PLANNER_RAW


Task completion and deadline success remain distinct
131-node graphs are outside routing-training sizes
On-time delivery (%)
Whole-horizon delivery (%)
Network nodes
Network nodes
NONE
ALL
FIXED_RX
KEEP_ONE
KEEP
LEARN
PLANNER_WARN
PLANNER_RAW